

Unit IV Activity II

Name: B C Yugesh

Reg no: 192424282

Course: Data Handling With Visualization

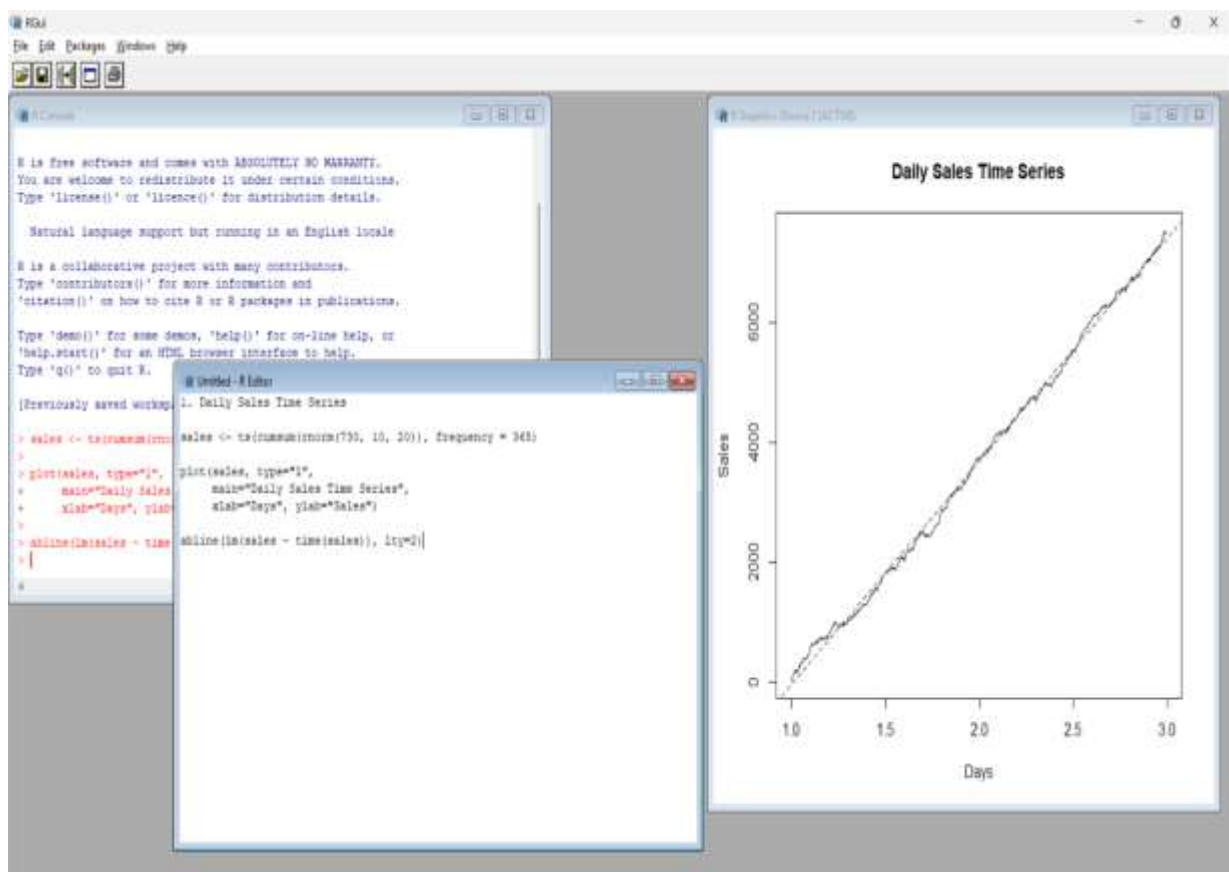
Course Code: DSA0604

1. Daily Sales Time Series

Program:

```
sales <- ts(cumsum(rnorm(730, 10, 20)), frequency = 365)
plot(sales, type="l",
     main="Daily Sales Time Series",
     xlab="Days", ylab="Sales")
abline(lm(sales ~ time(sales)), lty=2)
```

Output:

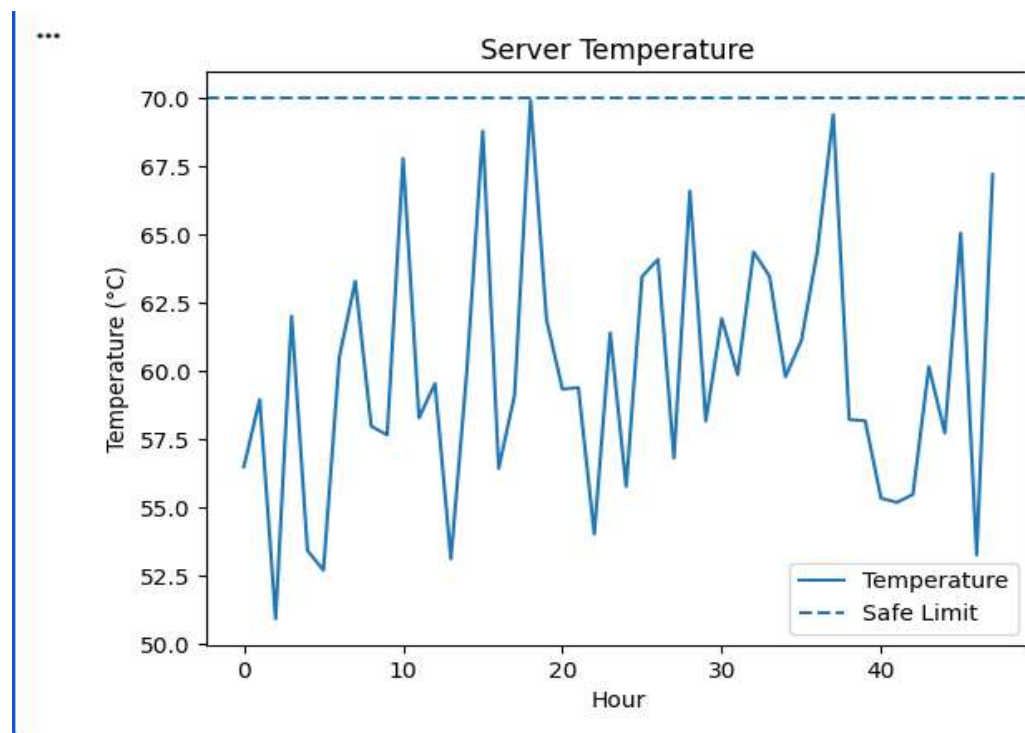


2. Server Temperature

Program:

```
import numpy as np
import matplotlib.pyplot as plt
temp = 60 + np.random.randn(48) * 5
threshold = 70
plt.plot(temp, label="Temperature")
plt.axhline(threshold, linestyle="--", label="Safe Limit")
plt.scatter(np.where(temp > threshold), temp[temp > threshold])
plt.xlabel("Hour")
plt.ylabel("Temperature (°C)")
plt.title("Server Temperature")
plt.legend()
plt.show()
```

Output:

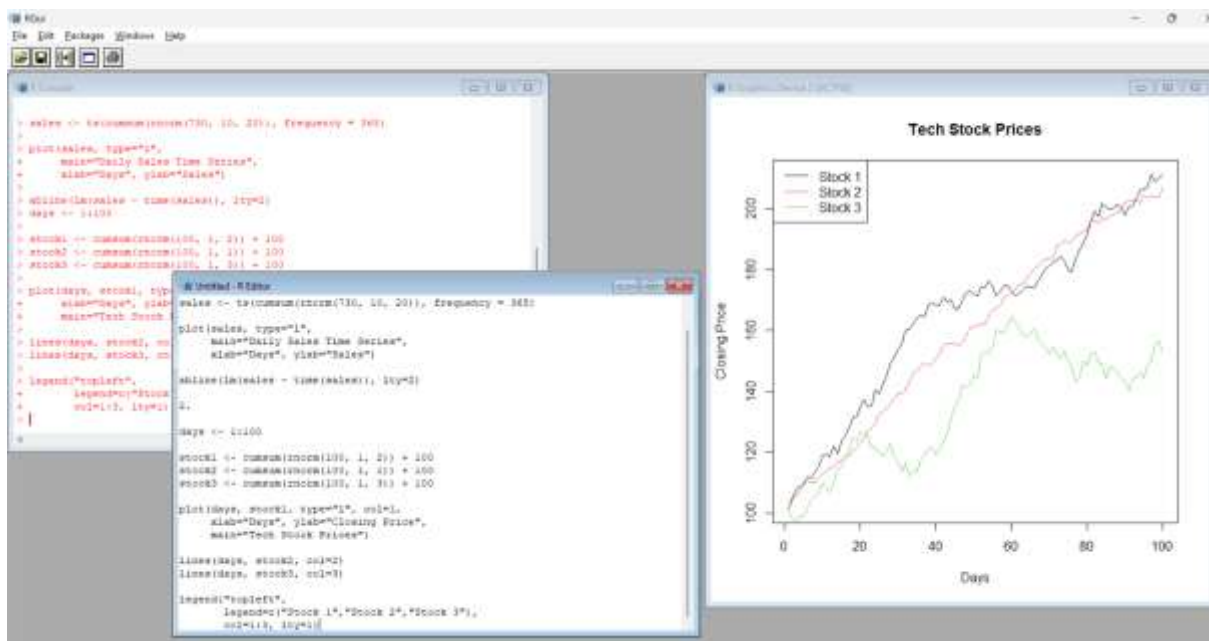


3. Three Tech Stocks

Program:

```
days <- 1:100  
  
stock1 <- cumsum(rnorm(100, 1, 2)) + 100  
stock2 <- cumsum(rnorm(100, 1, 1)) + 100  
stock3 <- cumsum(rnorm(100, 1, 3)) + 100  
  
plot(days, stock1, type="l", col=1,  
      xlab="Days", ylab="Closing Price",  
      main="Tech Stock Prices")  
  
lines(days, stock2, col=2)  
  
lines(days, stock3, col=3)  
  
legend("topleft",  
      legend=c("Stock 1", "Stock 2", "Stock 3"),  
      col=1:3, lty=1)
```

Output:

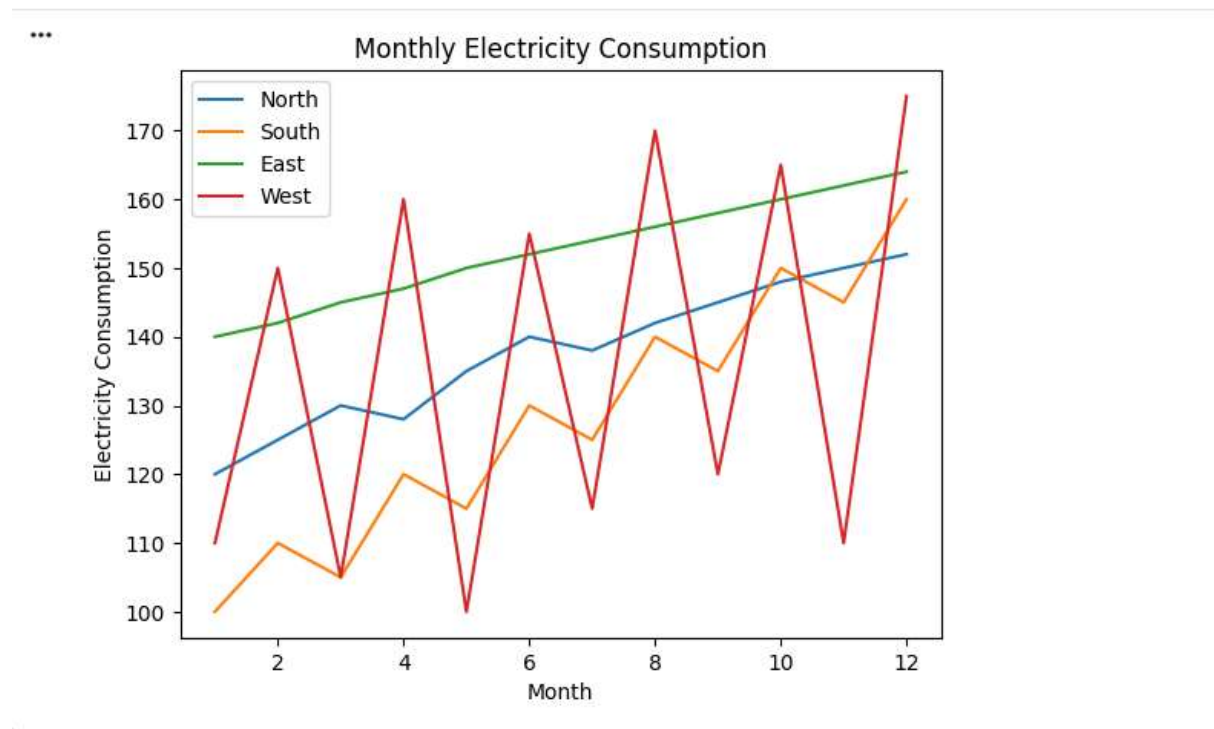


4. Electricity Consumption

Program:

```
import numpy as np
import matplotlib.pyplot as plt
months = np.arange(1, 13)
n1 = [120,125,130,128,135,140,138,142,145,148,150,152]
n2 = [100,110,105,120,115,130,125,140,135,150,145,160]
n3 = [140,142,145,147,150,152,154,156,158,160,162,164]
n4 = [110,150,105,160,100,155,115,170,120,165,110,175]
plt.plot(months, n1, label="North")
plt.plot(months, n2, label="South")
plt.plot(months, n3, label="East")
plt.plot(months, n4, label="West")
plt.xlabel("Month")
plt.ylabel("Electricity Consumption")
plt.title("Monthly Electricity Consumption")
plt.legend()
plt.show()
```

Output:

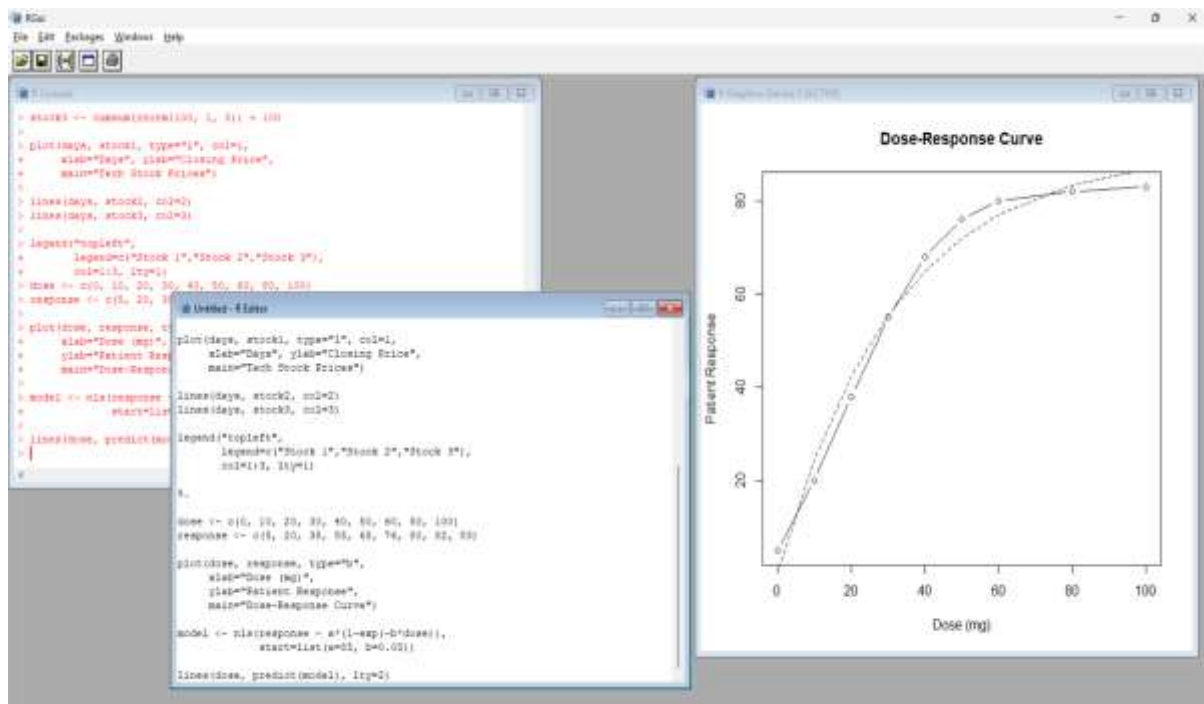


5. Drug Dose–Response Curve

Program:

```
dose <- c(0, 10, 20, 30, 40, 50, 60, 80, 100)
response <- c(5, 20, 38, 55, 68, 76, 80, 82, 83)
plot(dose, response, type="b",
     xlab="Dose (mg)",
     ylab="Patient Response",
     main="Dose-Response Curve")
model <- nls(response ~ a*(1-exp(-b*dose)),
             start=list(a=85, b=0.05))
lines(dose, predict(model), lty=2)
```

Output:



6. Fertilizer Dose–Response

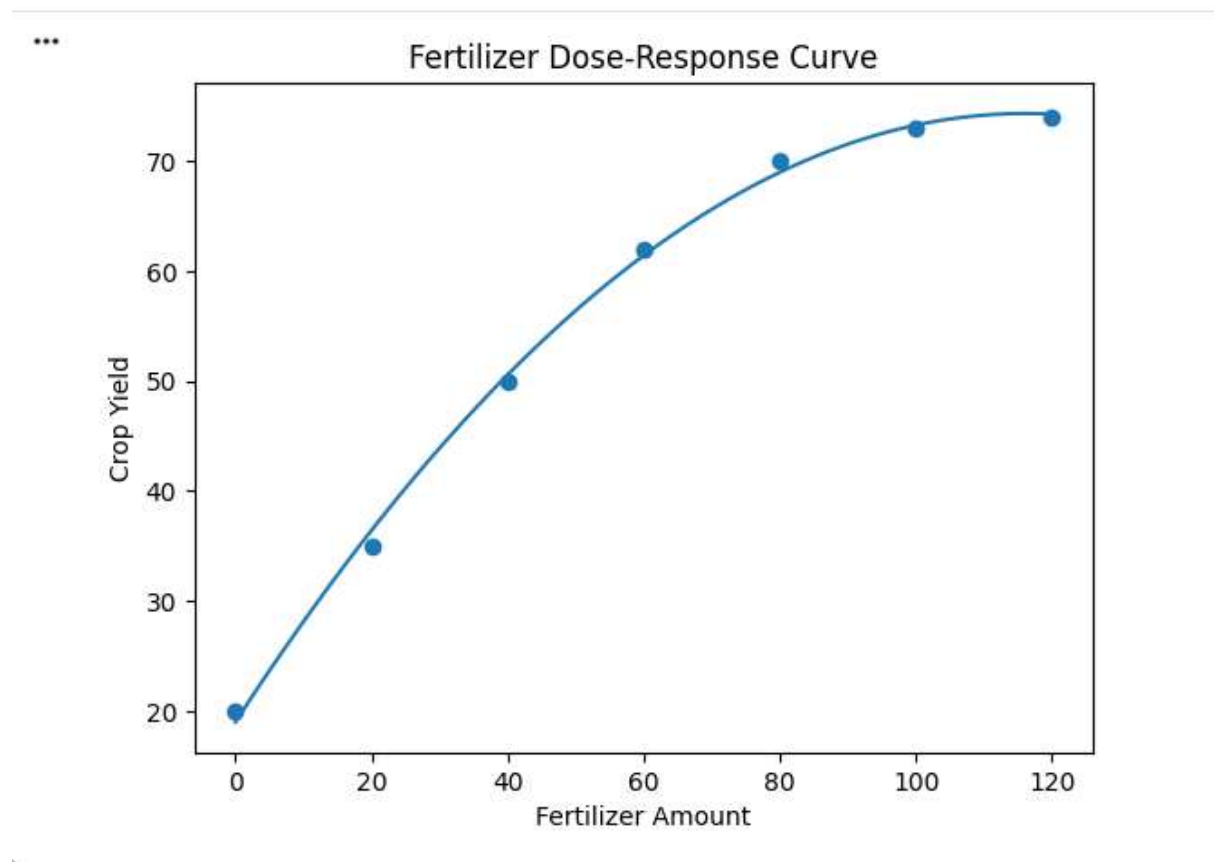
Program:

```
import numpy as np
import matplotlib.pyplot as plt

fertilizer = np.array([0,20,40,60,80,100,120])
yield_data = np.array([20,35,50,62,70,73,74])
coef = np.polyfit(fertilizer, yield_data, 2)
x = np.linspace(0,120,100)
y = np.polyval(coef, x)
plt.scatter(fertilizer, yield_data)
plt.plot(x, y)
plt.xlabel("Fertilizer Amount")
plt.ylabel("Crop Yield")
plt.title("Fertilizer Dose-Response Curve")
```

```
plt.show()
```

Output:

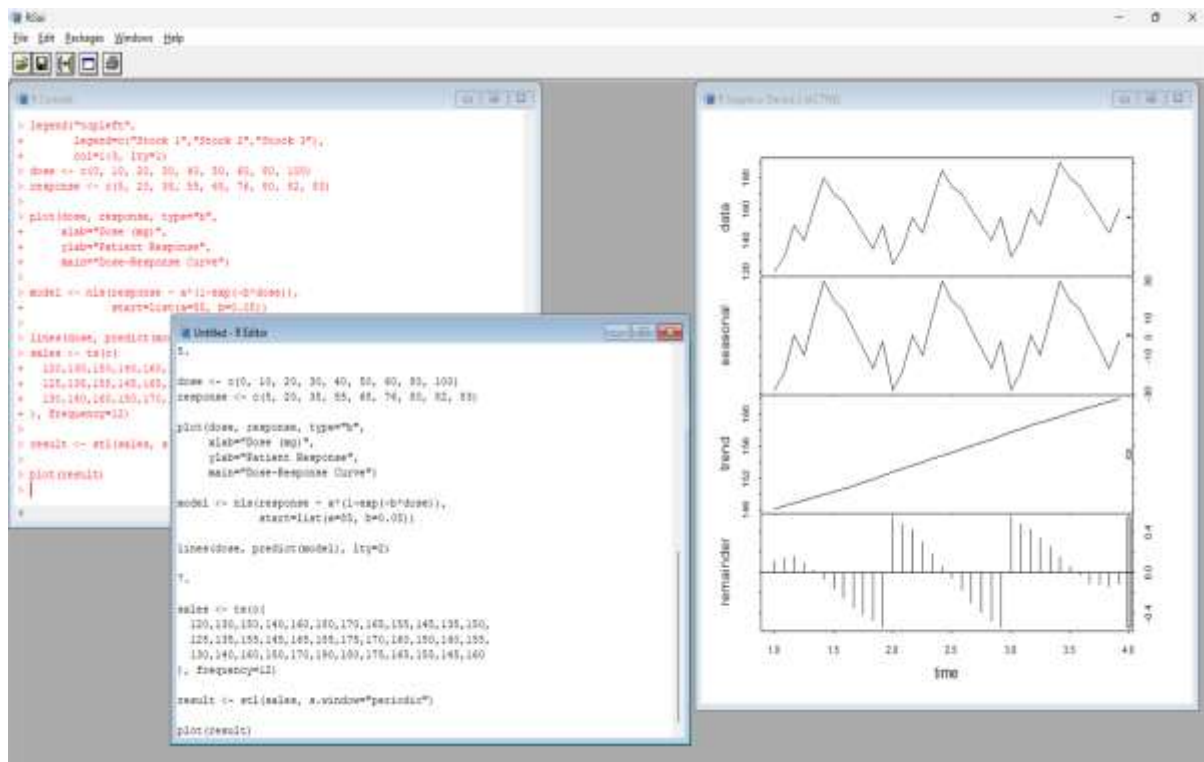


7. STL Decomposition of Ticket Sales

Program:

```
sales <- ts(c(
  120,130,150,140,160,180,170,165,155,145,135,150,
  125,135,155,145,165,185,175,170,160,150,140,155,
  130,140,160,150,170,190,180,175,165,155,145,160
), frequency=12)
result <- stl(sales, s.window="periodic")
plot(result)
```


Output:



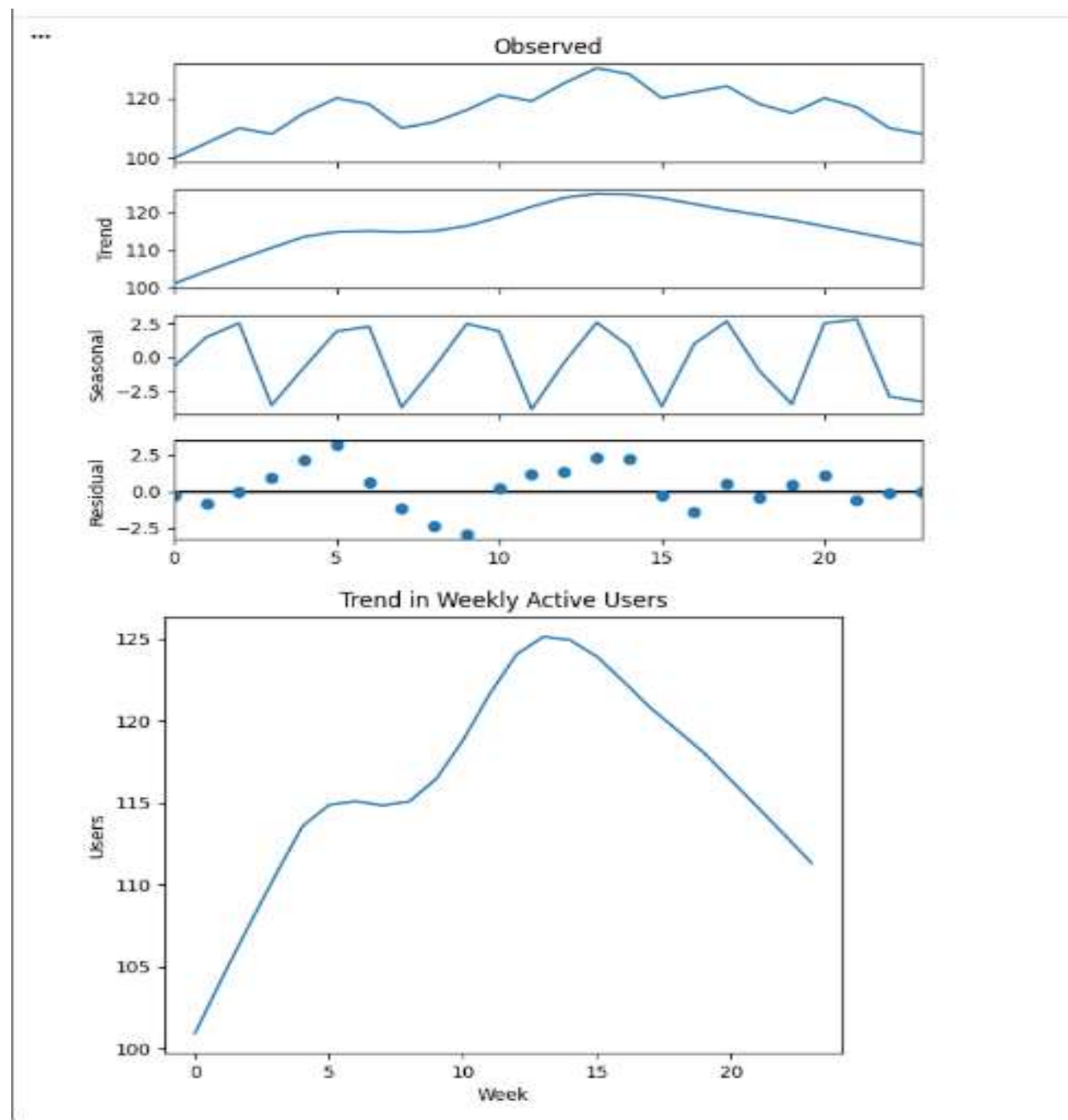
8. Weekly User Activity

Program:

```
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import STL
users = np.array([
    100,105,110,108,115,120,118,110,
    112,116,121,119,125,130,128,120,
    122,124,118,115,120,117,110,108
])
result = STL(users, period=4).fit()
result.plot()
plt.show()
```

```
plt.figure()
plt.plot(result.trend)
plt.title("Trend in Weekly Active Users")
plt.xlabel("Week")
plt.ylabel("Users")
plt.show()
```

Output:

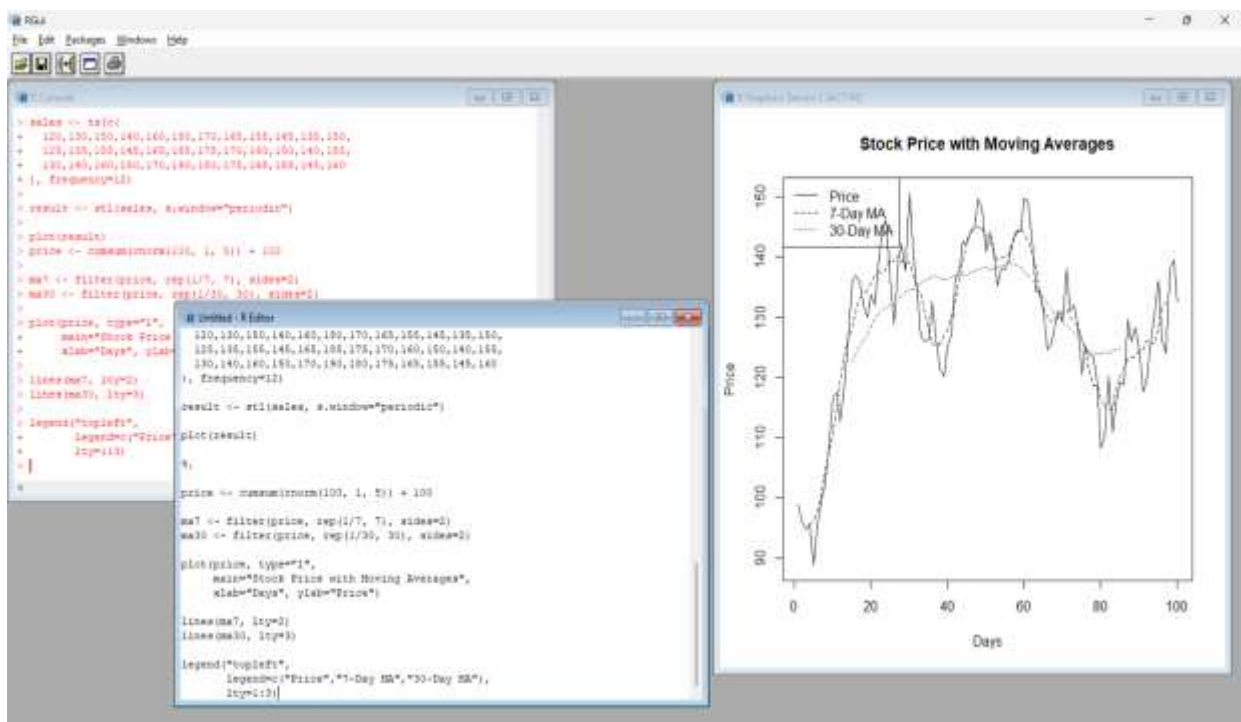


9. 7-Day and 30-Day Moving Average

Program:

```
price <- cumsum(rnorm(100, 1, 5)) + 100
ma7 <- filter(price, rep(1/7, 7), sides=2)
ma30 <- filter(price, rep(1/30, 30), sides=2)
plot(price, type="l",
      main="Stock Price with Moving Averages",
      xlab="Days", ylab="Price")
lines(ma7, lty=2)
lines(ma30, lty=3)
legend("topleft",
      legend=c("Price", "7-Day MA", "30-Day MA"),
      lty=1:3)
```

Output:



10. GDP Detrending

Program:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

gdp = np.array([
    500,510,515,525,520,535,545,550,
    548,560,570,575,580,590,600,610
])

detrended = signal.detrend(gdp)
plt.plot(detrended)
plt.axhline(0, linestyle="--")
plt.xlabel("Quarter")
plt.ylabel("Detrended GDP")
plt.title("Detrended GDP Series")
plt.show()
```

Output:

